

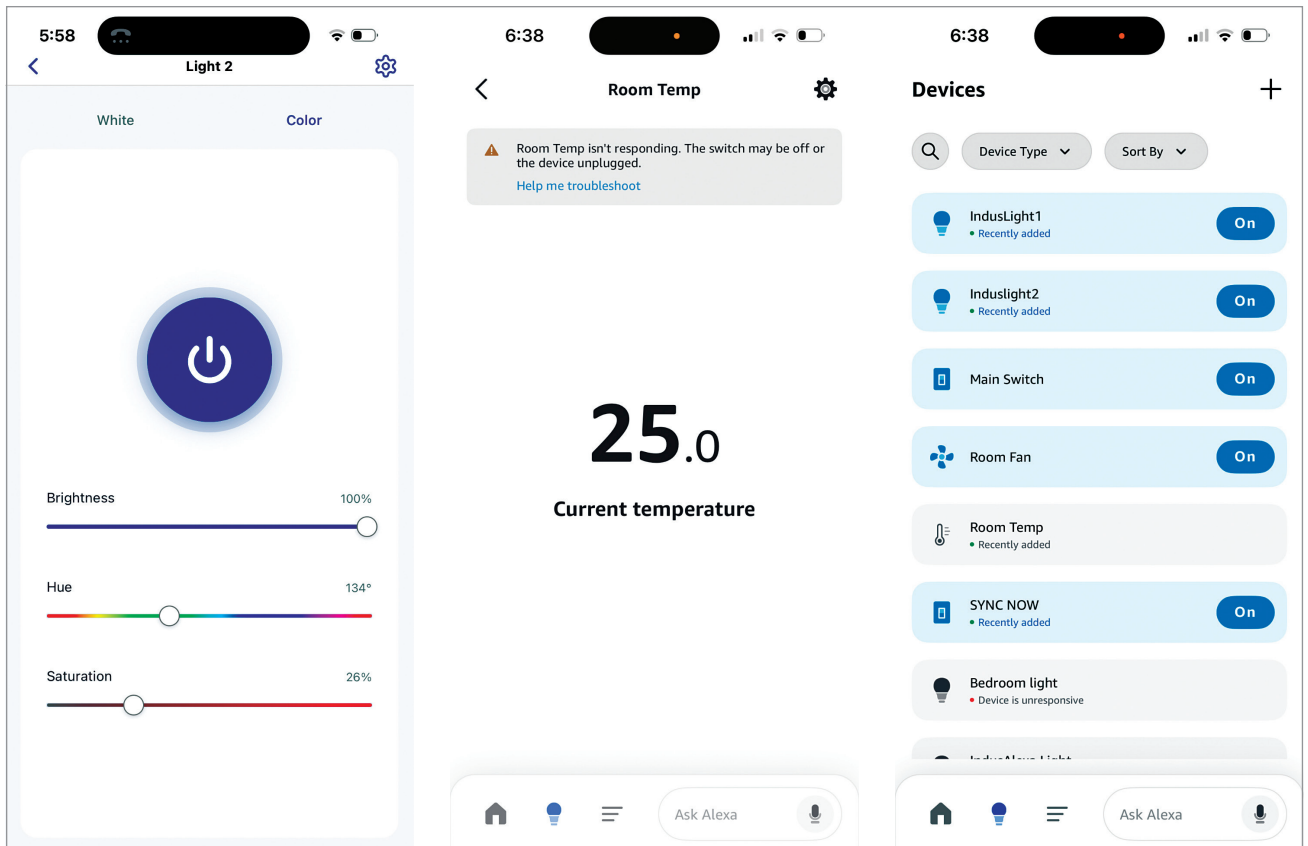


Fig. 11: RainMaker Home app (L) interface for colour and brightness control, and Alexa app (M) showing current room temperature and devices such as lights, fan, switch, and sensors (R)

Testing

After uploading the code and connecting the power supply, the system is ready for testing. This system functions as a production-ready platform suitable for mass deployment.

It provides a comprehensive dashboard, device insights, and a device management framework for clients. Using the RainMaker platform, devices can be monitored, crash reports reviewed, and firmware updates delivered remotely.

Log in to the RainMaker dashboard at <https://dashboard.rain-maker.espressif.com> to view all registered devices, their status, and available insights. It is possible to track how many devices are online, operating correctly, requiring restarts, or experiencing software issues. These insights support effective technical support and over-the-air (OTA) firmware updates.

Fig. 9 shows the RainMaker dashboard displaying device insights, while Fig. 10 shows firmware management and OTA updates.

End users or system integrator clients can control the devices using the RainMaker Home app or a custom client application. Elements such as lights, fans, actuators, and sensors are automatically displayed in the app interface based on the firmware configuration.

This system includes full RGB pixel LED control with 16 million

colours, two switches, and a fan with speed control and temperature sensing. All parameters can be monitored and controlled through the app interface. Light colour, brightness, and on/off state can be adjusted through the app, and the same controls are available via the Alexa app or voice commands. For example, saying, “Alexa, turn the room light to green or warm white and set the brightness to 80 per cent,” performs the requested action. The fan can also be controlled through the app or voice commands.

Fig. 11 shows the RainMaker Home app interface for light colour and brightness control, alongside the Alexa app displaying current room temperature and device control options. **EFY**

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